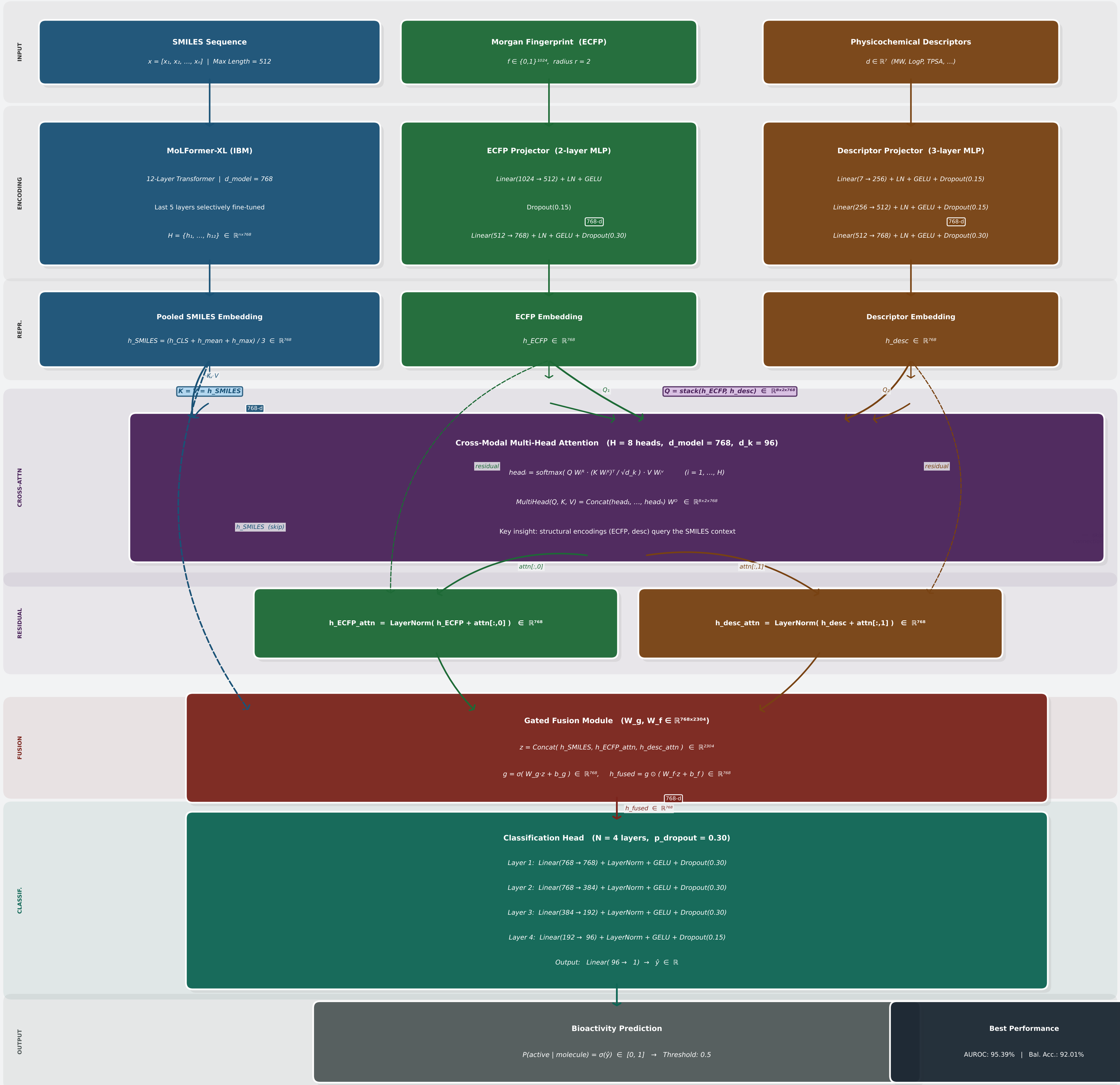
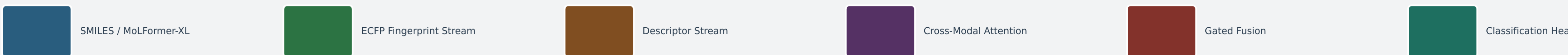


FineTunedBERTaECFP: Multimodal Bioactivity Prediction Architecture

MolFormer-XL (SMILES) ⊗ ECFP Fingerprint ⊗ Physicochemical Descriptors → Binary Bioactivity



Module Colour Legend



Training: RMSprop + OneCycleLR $\mid$ Regularisation: Dropout(0.30), Weight Decay(1.24×10^{-4}), Grad. Clipping(1.0), Grad. Accum.(x2) $\mid$ Optuna TPE (50 trials)